



INDIA'S THORIUM BET

INDIA'S PROTOTYPE FAST BREEDER REACTOR HAS ACHIEVED CRITICALITY, ADVANCING THE THREE-STAGE NUCLEAR STRATEGY. IS THE COUNTRY CLOSE TO USING ITS VAST THORIUM RESERVES TO BECOME ATMANIRBHAR IN NUCLEAR POWER GENERATION?

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IMAGE BY **BANDEEP SINGH/AI**

After 20 years of work, fast breeder reactor (FBR) achieved criticality in Kalpakkam last month. India is the second country after Russia to have the FBR. This needs to be scaled up to produce Uranium 233, which is not available in nature, for Stage 3

REACTOR TYPE



Fast Breeder

FUEL IN / OUT

Plutonium-239 + Thorium-232 Uranium-233

STRATEGIC PURPOSE

Multiply fissile inventory; introduce thorium



Exploit vast thorium reserves with Uranium 233 in Advanced Heavy Water Reactors. The stage is still decades away as a large amount of U233 is required from Stage 2 to fuel the reactors

REACTOR TYPE



Advanced Heavy Water

FUEL IN / OUT

Uranium-233 + Thorium-232 Power

STRATEGIC PURPOSE

Exploit vast thorium reserves at scale